

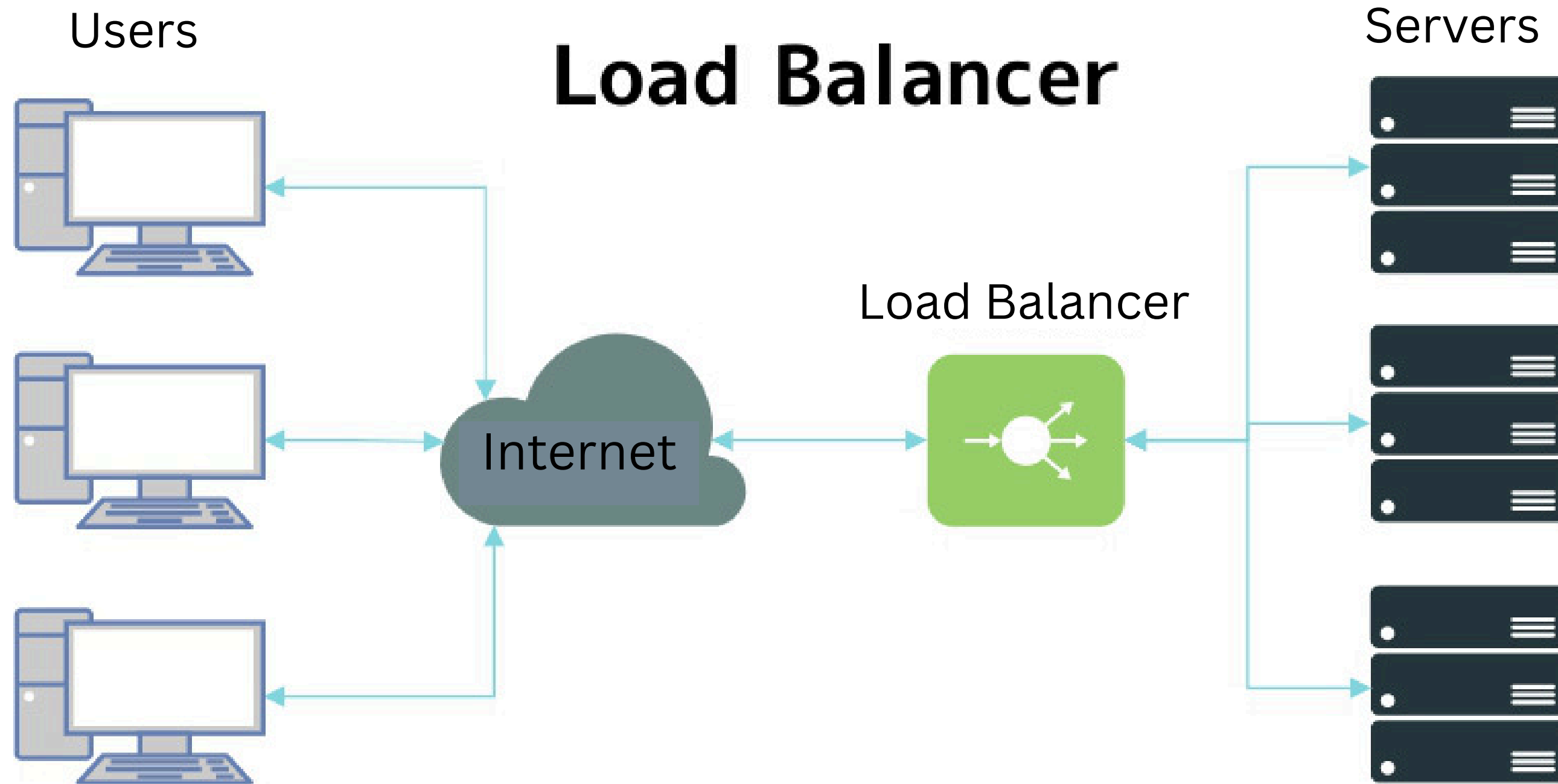
Addressing the World IPv4, Classful History, and CIDR

Computer Networks

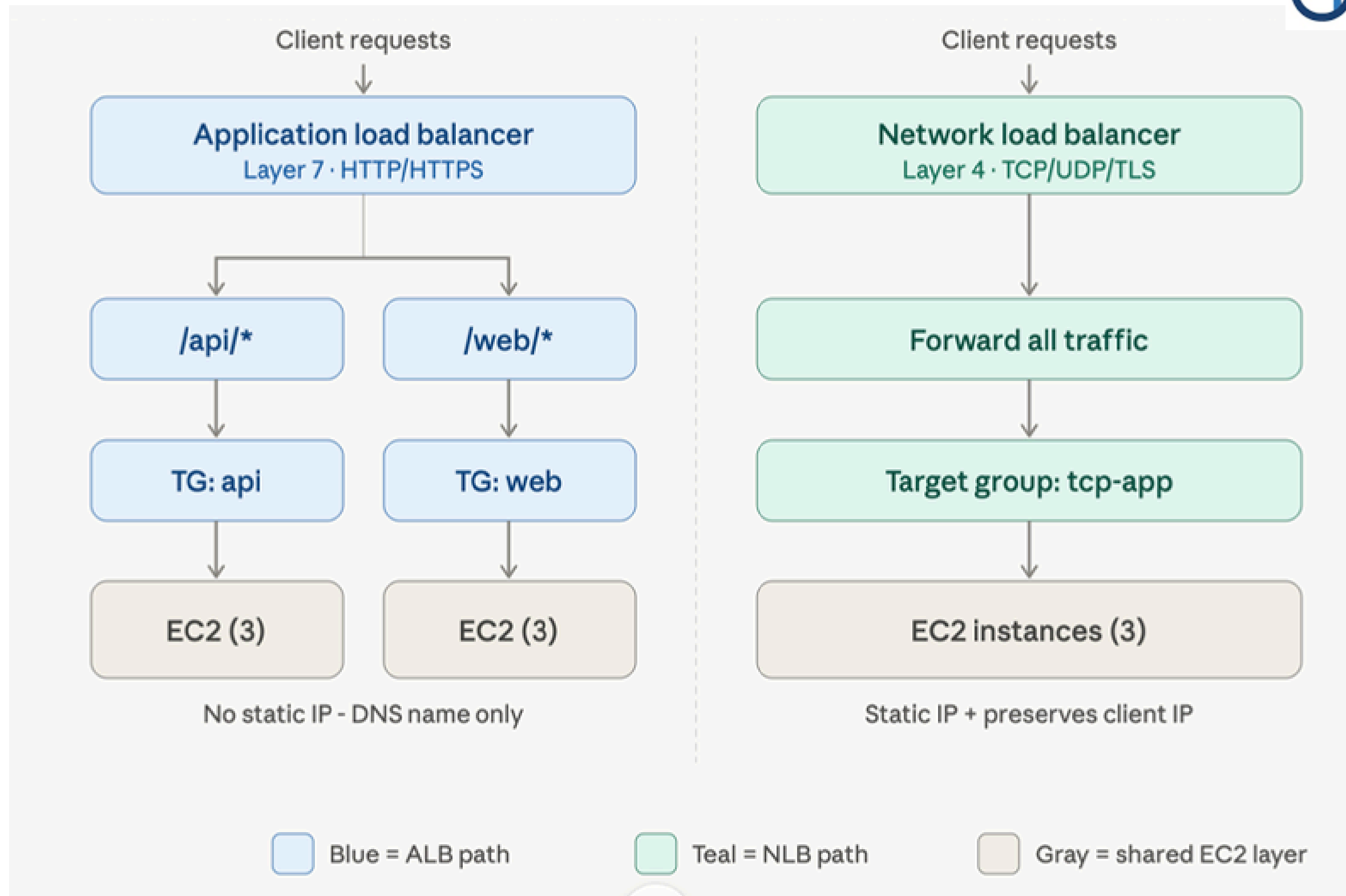
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Network Layer: IP Addressing

Recap

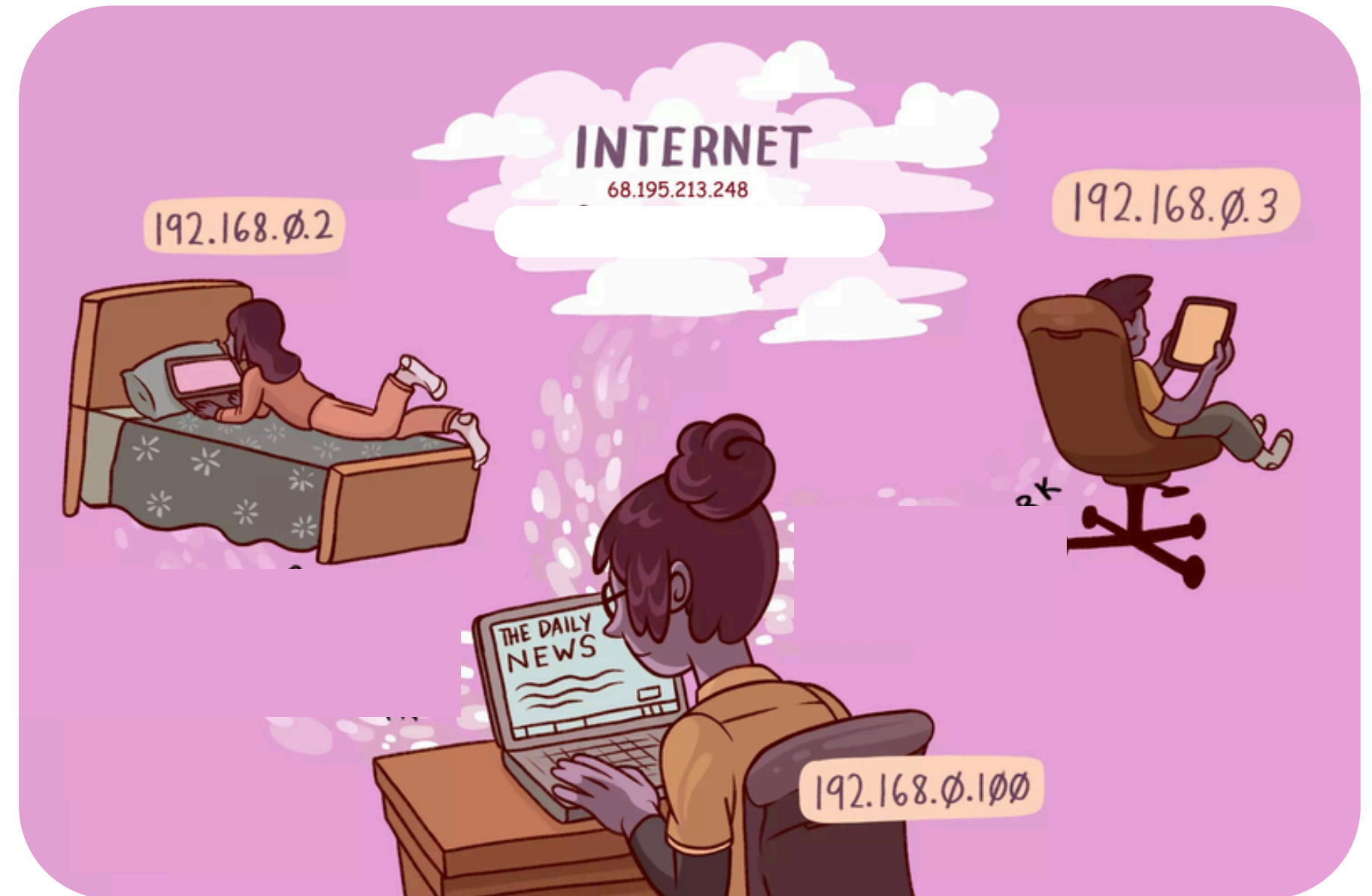


Recap



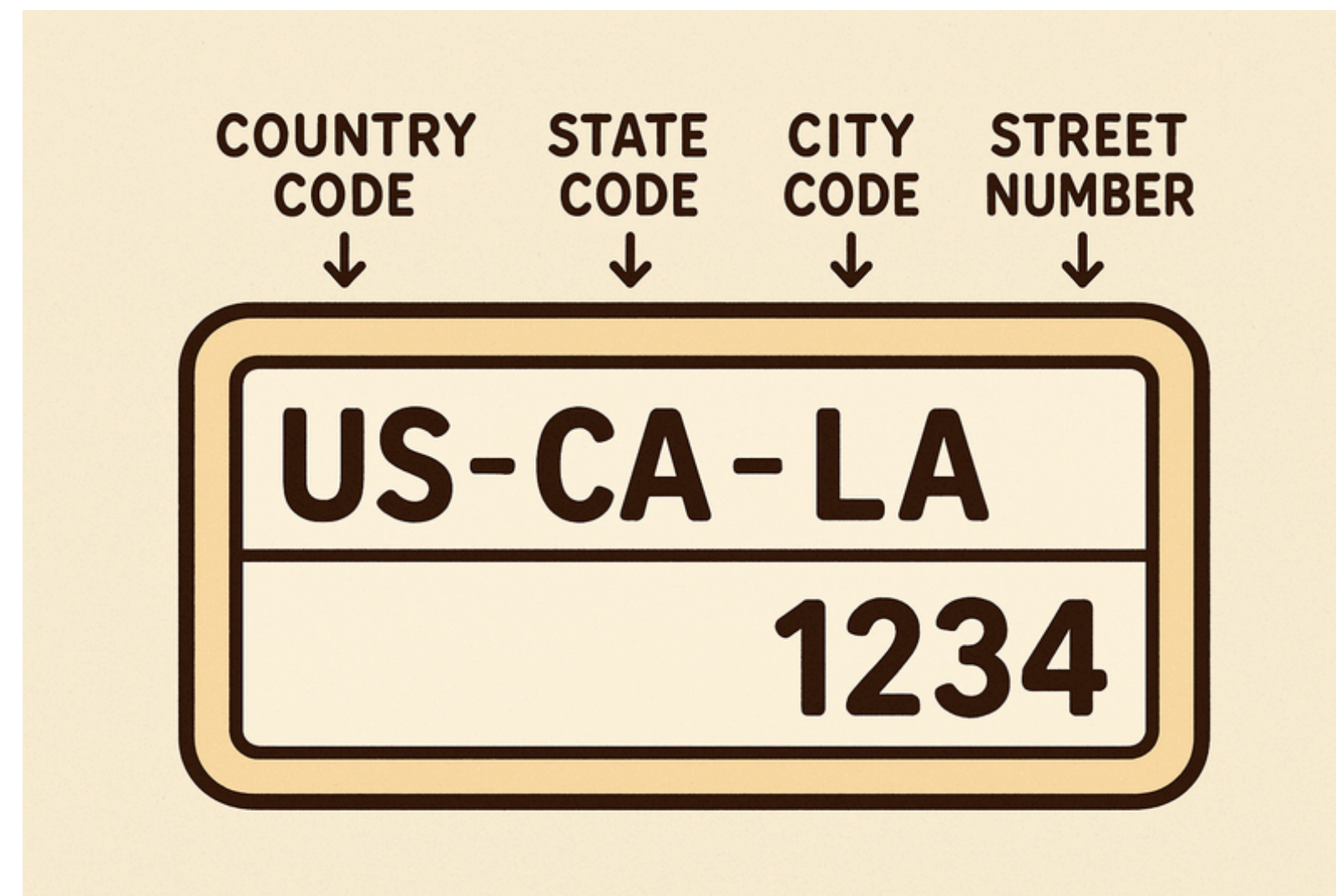
IP Address

An IP address is a unique identifier assigned to each device connected to the internet or a local network. "IP" stands for "Internet Protocol," the set of rules for sending data between devices on networks.



IPv4 Addressing

*Think of the IPv4 address as a simple, four-part postal code:
[Country Code].[State Code].[City Code].[Street Number].*



IPv4 Addressing

- The first major version, which still powers most of the internet today.
- **Structure:** A 32-bit number.
- **Format:** Represented as four 8-bit numbers (octets), separated by dots.
- **Example:** 192.168.1.1
- Each of the four numbers can range from 0 to 255.



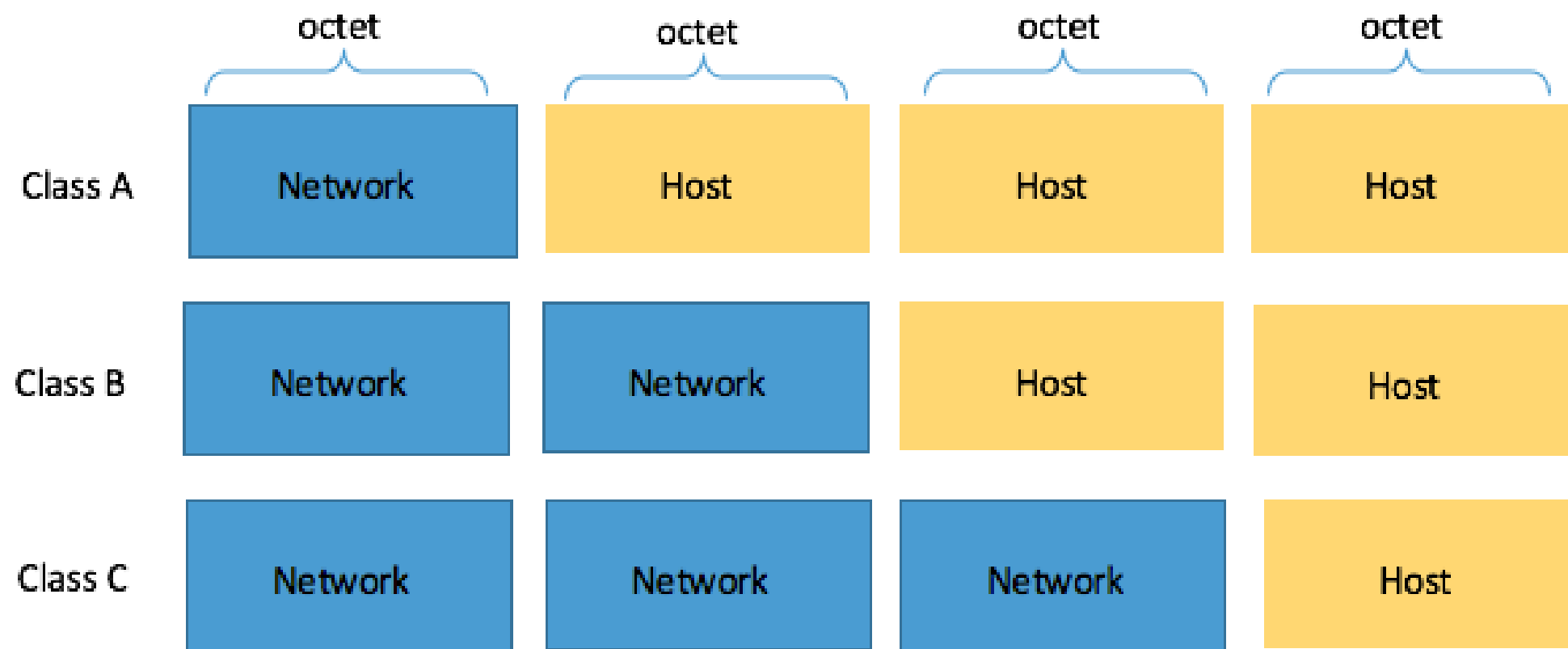
History of IPv4 address management

History of IPv4 Address Classes

"The city model is divided into three structured zones:
A (Massive districts for giant developments), B (Medium districts for mid-sized projects), and C (Small plots for local use)."



IPv4 Address Classes A, B and C



172.16.1.10

Network ID Host ID

Class	Left-most bit	Starting IP address	Last IP address
A	0xxx	0.0.0.0	127.255.255.255
B	10xx	128.0.0.0	191.255.255.255
C	110x	192.0.0.0	223.255.255.255

Classful IPv4 addresses

"The classes had a significant amount of difference in available IP addresses for hosts. This meant practically a lot of IPv4 addresses remained unassigned"

Class	Bit Allocation (32 bits total)
A	0NNNNNNN.HHHHHHHH.HHHHHHHH.HHHHHHHH (1 bit class ID + 7 network bits + 24 host bits)
	Range: 0.0.0.0 to 127.255.255.255 Networks: ~128 Hosts per network: ~16.7 million
B	10NNNNNNN.NNNNNNNN.HHHHHHHH.HHHHHHHH (2 bits class ID + 14 network bits + 16 host bits)
	Range: 128.0.0.0 to 191.255.255.255 Networks: ~16,000 Hosts per network: ~65,000
C	110NNNNN.NNNNNNNN.NNNNNNNN.HHHHHHHH (3 bits class ID + 21 network bits + 8 host bits)
	Range: 192.0.0.0 to 223.255.255.255 Networks: ~2 million Hosts per network: 254

Think about how many potential devices in a flat Class A network?

IP address classes

Class	Range	Max Hosts
A	0-127	16 Million
B	128-191	64,000
C	192-223	254

For class A and Class B, that's too many devices to be managed by a single router!

The Journey of a Packet to 9.10.10.10

Step 1: Core Router Recognition

Core routers identify 9.10.10.10 belongs to 9.0.0.0 Class A network

Route packet using network ID to appropriate gateway

Step 2: Gateway Router Processing

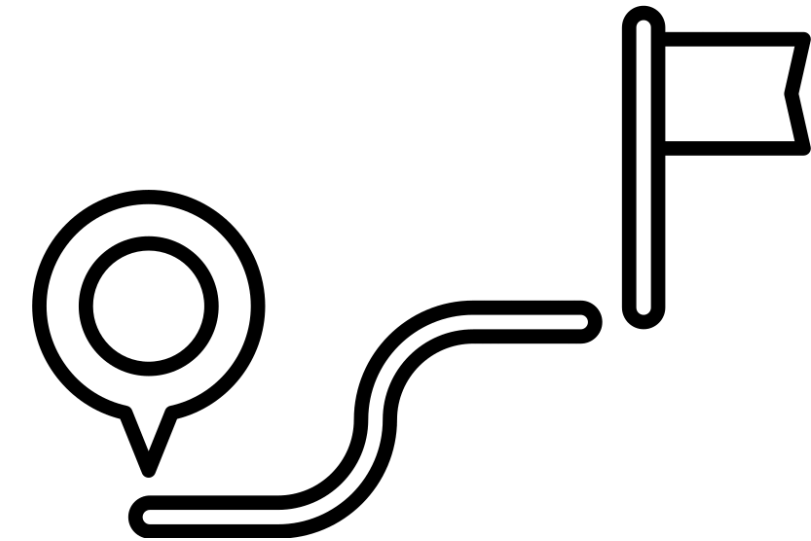
Gateway router = entry/exit point for specific network

Uses host ID to deliver to final destination

The Scale Problem!

16,777,216 individual IPs in a single Class A network

Way too many devices for one router!



IP Address Lookup for 9.10.10.10 in Durham, United States

Location: Durham, United States - 9.10.10.10 is an IP address allocated to IBM.
[Learn more.](#)

 WhatIsMyIPAddress

The Crisis: The Great Address Shortage!

- The internet grew much faster than anyone predicted!
- A company needing 500 addresses couldn't get a Class C (too small) and had to take a Class B (with over 65,000 addresses), wasting the rest.
- We were running out of available IPv4 addresses.
- In 2025, purchasing a public IPv4 address costs roughly \$20–\$45 per address, while leasing can be done for about \$0.40 per IP per month.



Solution: Make the routing classless!

CIDR: Classless Inter-Domain Routing

Replaces the older Class A, B, C system

- Networks can be created based on actual host requirements instead of wasting large address blocks.
- CIDR makes it easier to divide a network into smaller subnets as per need

CIDR Format

192.168.0.1/24

Normal IP + "/" + Network Prefix

- 192.168.0.1 = Network address
- /24 = Network prefix (subnet mask)

CIDR Value & CIDR Notation

IP Address: 192.168.0.1
Subnet Mask: 255.255.255.0

Binary Subnet Mask
11111111.11111111.11111111.00000000



CIDR Notation = IP Address/CIDR Value

CIDR Notation = 192.168.0.1/24

CIDR

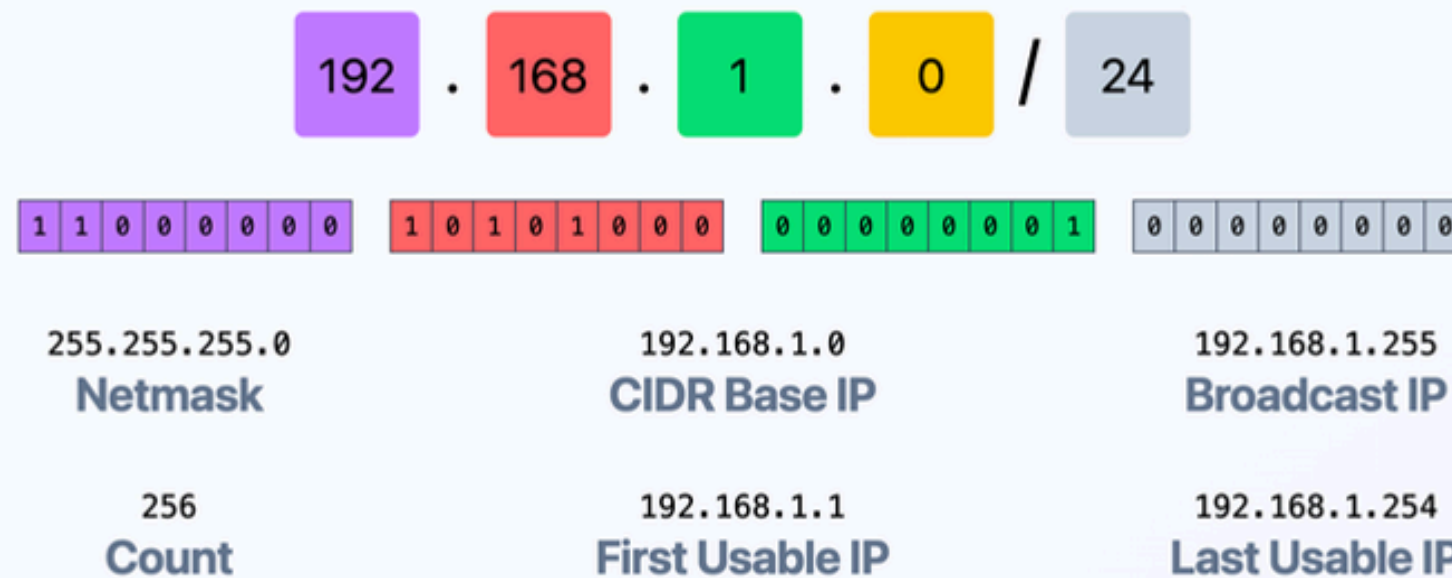
Network = 24 bits

Host = 8 bits

11111111 . 11111111 . 11111111 . 00000000

255 . 255 . 255 . 0

/24 (Number of 1's)



Usable Hosts

$2^h - 2$ $h = \text{host bits}$

CIDR Base IP

Broadcast IP

192.168.1.0/24

Usable Hosts = $2^8 - 2 = 254$

CIDR Examples

CIDR Notation	Network Address	Subnet Mask	First Usable IP	Last Usable IP	Total Hosts	Use Case
10.0.0.0/8	10.0.0.0	255.0.0.0	10.0.0.1	10.255.255.254	16,777,214	Large enterprise networks
172.16.0.0/12	172.16.0.0	255.240.0.0	172.16.0.1	172.31.255.254	1,048,574	Medium enterprise networks
192.168.0.0/16	192.168.0.0	255.255.0.0	192.168.0.1	192.168.255.254	65,534	Large home/office networks
192.168.1.0/24	192.168.1.0	255.255.255.0	192.168.1.1	192.168.1.254	254	Standard home/small office
192.168.1.0/25	192.168.1.0	255.255.255.128	192.168.1.1	192.168.1.126	126	Small office subnets
192.168.1.0/26	192.168.1.0	255.255.255.192	192.168.1.1	192.168.1.62	62	Department networks
192.168.1.0/27	192.168.1.0	255.255.255.224	192.168.1.1	192.168.1.30	30	Small team networks
192.168.1.100/32	192.168.1.100	255.255.255.255	192.168.1.100	192.168.1.100	1	Single host/loopback

This means you could request an address block of exactly the size you needed, dramatically reducing waste.

We need at least 500 IP addresses to connect 500 devices in one subnet

1 500 devices

500



3 Take 9 bits for hosts

Network · 23 bits

Hosts · 9 bits

9 host bits → 2⁹ = 512 total addresses. 512 – 2 (network and broadcast) = 510 usable IPs — enough for 500 devices.

5 Summary

Total bits
32

Network bits
23

Host bits
9

Usable IPs
510

Supports
500 devices

2 Need at least 500 usable IPs

Find the smallest host bits H so that 2^H – 2 ≥ 500.

Host bits (H)	Usable IPs (2 ^H – 2)
7	126
8	254
9	510

Enough

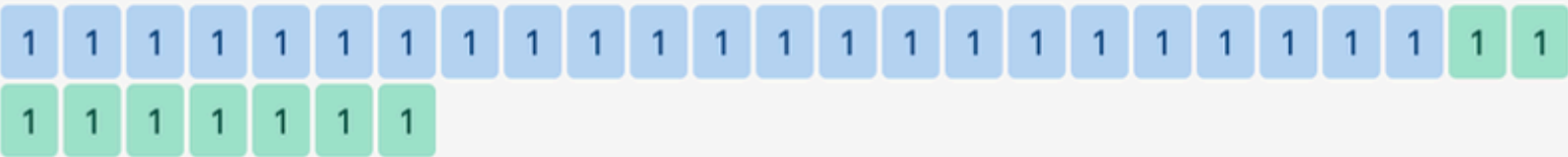
4 Example: using 192.168.1.0/24

× Not enough

A /24 has 32 – 24 = 8 host bits → only 254 usable IPs

✓ New prefix: /23

Borrow 1 bit → 32 – 23 = 9 host bits → 510 usable IPs



23 bits network · 9 bits hosts

Traditional network classes and their CIDRs

Class	Default Subnet Mask with CIDR representations
A	255.0.0.0 (/8)
B	255.255.0.0 (/16)
C	255.255.255.0 (/24)

Private Addresses: Reusable not Public

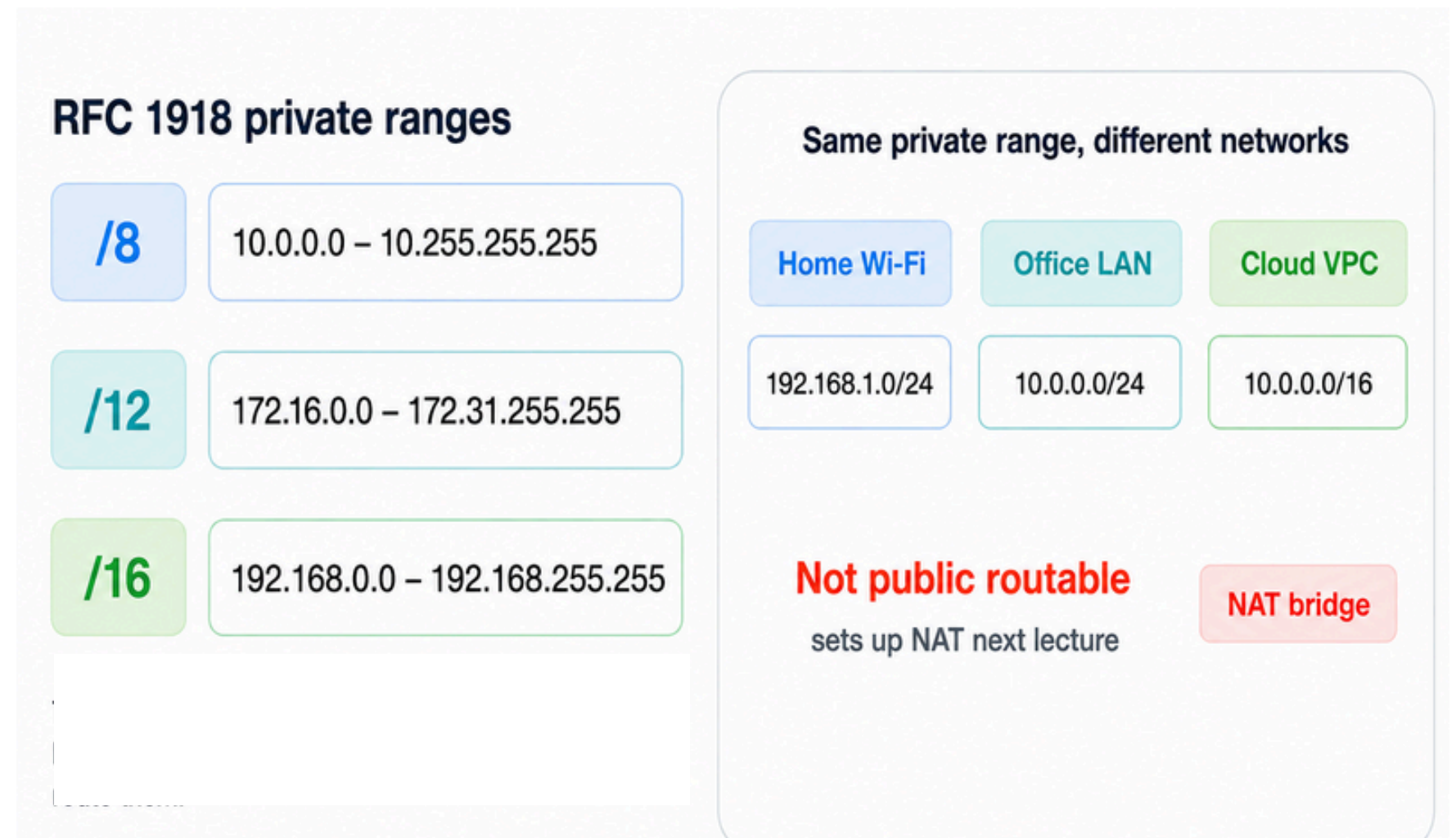
Private IP addresses are utilized in local networks like homes and offices.

The same private IP range can be reused across different networks.

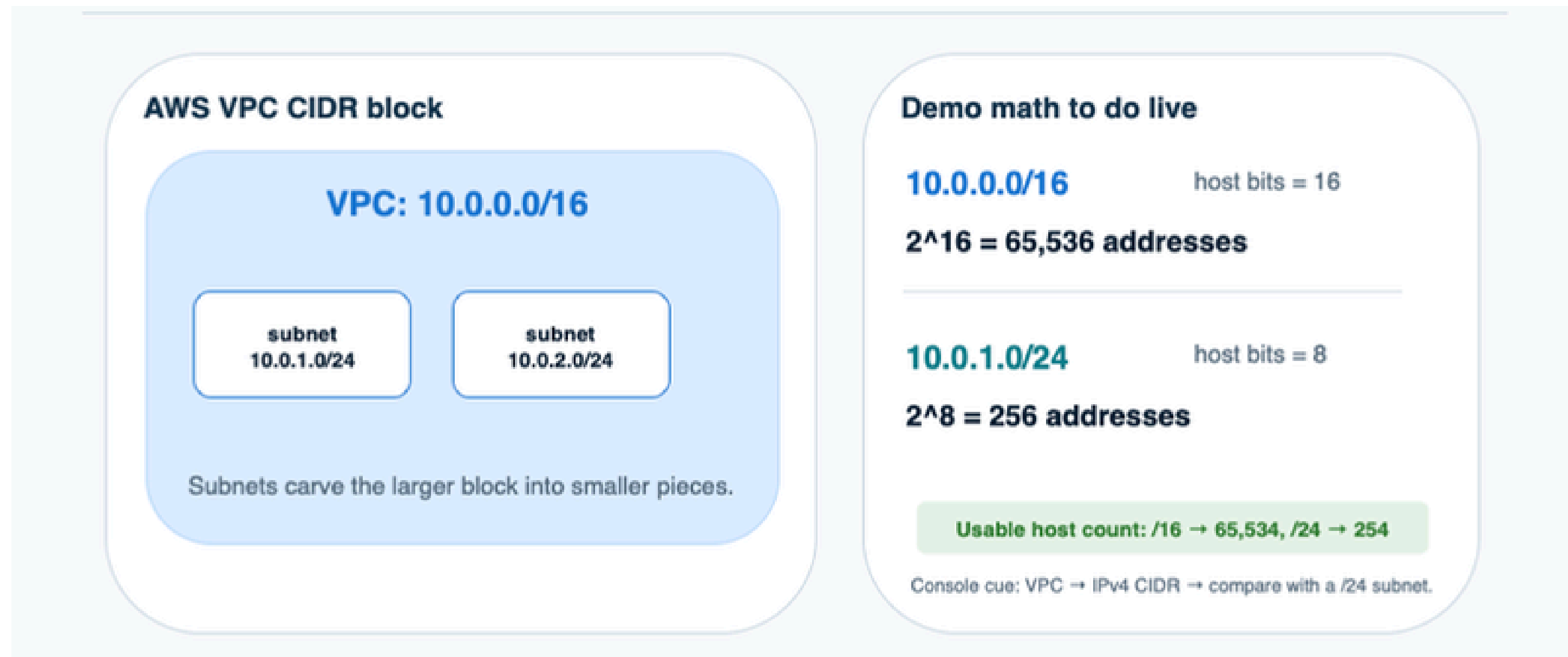
For instance:

- Your home laptop might have 192.168.1.10.
- Another student's laptop in a different city could also have 192.168.1.10.

Both are valid as they exist in separate private networks.



Cloud bridge: VPCs use the same CIDR math



Cloud bridge: VPCs use the same CIDR math

VPCs

VPC encryption controls

Your VPCs (1/2)

Info

Last updated 1 minute ago

Actions

Create VPC

Find VPCs by attribute or tag

	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public..
☒	CN-Lab-VPC-vpc	vpc-08062d5ccf8104af0	Available	-	-	Off
☐	-	vpc-0f34a71fce73316ff	Available	-	-	Off

vpc-08062d5ccf8104af0 / CN-Lab-VPC-vpc

DetailsResource mapCIDRFlow logsTagsIntegrations

IPv4 CIDRs

Info

Address familyCIDRStatus

IPv4	10.0.0.0/16	Associated
------	-------------	------------

Edit CIDRs

Subnets (1/10)

Info

Last updated 1 minute ago

Actions

Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IP
☒	CN-Lab-VPC-subnet-private2-us-east-1b	subnet-0751527e050159830	Available	vpc-08062d5ccf8104af0 CN-L...	Off	10.0.0.0/20
☐	CN-Lab-VPC-subnet-public1-us-east-1a	subnet-05d488676845659e3	Available	vpc-08062d5ccf8104af0 CN-L...	Off	10.0.0.0/20
☐	-	subnet-0db376f5974b502b7	Available	vpc-0f34a71fce73316ff	Off	10.0.0.0/20
☐	-	subnet-0921c3e5132120302	Available	vpc-0f34a71fce73316ff	Off	10.0.0.0/20
☐	-	subnet-0a178c5d3700e46d8	Available	vpc-0f34a71fce73316ff	Off	10.0.0.0/20

subnet-0751527e050159830 / CN-Lab-VPC-subnet-private2-us-east-1b

Details

Subnet ID

Subnet ARN

IPv4 CIDR

Availability Zone

State

Block Public Access

IPv6 CIDR

IPv6 CIDR association ID

VPC

Route table

Subnets (1/10)

Info

Last updated 2 minutes ago

Actions

Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IP
☐	CN-Lab-VPC-subnet-private2-us-east-1b	subnet-0751527e050159830	Available	vpc-08062d5ccf8104af0 CN-L...	Off	10.0.0.0/20
☒	CN-Lab-VPC-subnet-public1-us-east-1a	subnet-05d488676845659e3	Available	vpc-08062d5ccf8104af0 CN-L...	Off	10.0.0.0/20
☐	-	subnet-0db376f5974b502b7	Available	vpc-0f34a71fce73316ff	Off	10.0.0.0/20
☐	-	subnet-0921c3e5132120302	Available	vpc-0f34a71fce73316ff	Off	10.0.0.0/20
☐	-	subnet-0a178c5d3700e46d8	Available	vpc-0f34a71fce73316ff	Off	10.0.0.0/20

subnet-05d488676845659e3 / CN-Lab-VPC-subnet-public1-us-east-1a

Details

Subnet ID

Subnet ARN

IPv4 CIDR

Availability Zone

State

Block Public Access

IPv6 CIDR

IPv6 CIDR association ID

VPC

Route table

Please fill the feedback form.

Thank You